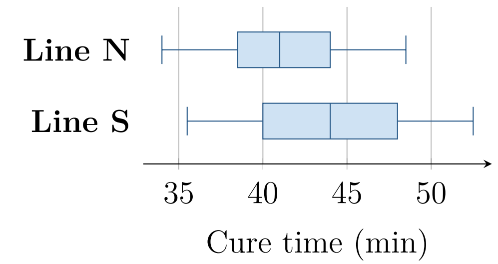
Two Test Plans for Cure Time

Topic 7.8 · TODAY'S PROBLEM

Suppose existing lines N (new formula) and S (standard) make the same sealant. Independently, a team samples 18 of 600 recent N batches and 18 of 540 recent S batches. Cure-time summaries are $\bar{x}_N = 41.2$, $s_N = 3.6$, $\bar{x}_S = 44.0$, and $s_S = 4.1$; boxplots are shown. Before sampling it asked, "Do the two populations differ in mean cure time?" Formula was not assigned to lines.



Rhea: "Preserve the pre-data direction, use population means, and separate association from causation."

Sol: "Since $41.2 < 44.0$, use a lower tail with sample means; significance proves formula caused faster curing."

FIRST TAKE (5 min, on your sheet)

Which parts of each plan do the question's timing and the design support? Name one detail to verify.

DOK 1 **(a)** Identify the response, both samples, and $\bar{x}_N - \bar{x}_S$. Did the pre-data question specify a direction?

DOK 2 **(b)** Name the procedure; define both parameters; state H_0 in N-minus-S order; and verify independence, 10 percent, and shape.

DOK 3 **(c)** Adjudicate both plans. Defend the alternative and notation, diagnose selecting a tail from the observed order, and bound a significant result. Explain the misleading belief invited by each rejected choice.

Video + follow-along: `u7_lesson8_live.html` (scan the code)

Finish (a)–(c) after the video. Turn in your sheet.